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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

MTH 101: Elementary Mathematics I - Algebra, Trigonometry and Sequences

Close the secondary-school gaps that make first-year mathematics feel impossible

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Mathematics | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

MTH 101 assumes you already own certain algebraic habits. Many freshers do not, and the course moves fast. This pack rebuilds those habits - factorisation, indices, quadratic theory, trigonometric identities and sequences - with short explanations and worked examples that model good layout.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Factorise and simplify algebraic expressions including surds and partial fractions.
- Solve quadratic equations and use the sum and product of roots.
- Apply trigonometric ratios, identities and the sine and cosine rules.
- Work confidently with arithmetic and geometric progressions, including sums to n terms.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Algebraic fluency

Speed and accuracy in algebra decide your grade more than clever tricks do.

- Know the difference of two squares, perfect squares and the factor theorem.
- Rationalise surds by multiplying by the conjugate.
- Solve simultaneous equations by substitution when one equation is linear.
- For quadratics, use factorisation first, completing the square second and the formula last.
- For ax^2 + bx + c = 0, the sum of roots is -b/a and the product is c/a.
- Check answers by substitution - it takes seconds and catches sign errors.

> **Show your line of reasoning**
> Marks are awarded for correct intermediate steps. A bare answer that is wrong earns nothing; a clear method with a slip often earns most of the marks.

Trigonometry that stays consistent

Trigonometry rewards a diagram and punishes memorised ratios applied blindly.

- Use the unit circle to determine the sign of each ratio in each quadrant.
- Sine rule: a/sin A = b/sin B = c/sin C.
- Cosine rule: a^2 = b^2 + c^2 - 2bc cos A.
- Area of a triangle: half ab sin C.
- Core identities: sin^2 + cos^2 = 1; tan = sin/cos; double-angle formulae.
- Convert between degrees and radians before using calculus formulae.

Sequences and series

Identify the type of sequence before touching a formula.

- Arithmetic progression: common difference d; nth term a + (n-1)d; sum n/2(2a + (n-1)d).
- Geometric progression: common ratio r; nth term ar^(n-1); sum a(1 - r^n)/(1 - r) for r not equal to 1.
- Sum to infinity exists only when the absolute value of r is less than 1, and equals a/(1 - r).
- For word problems, write the first three terms explicitly to confirm the pattern.
- Binomial expansion: use Pascal's triangle for small powers, the general term for larger ones.

Sets, logic and proof

Set notation is the grammar of mathematics: union, intersection, complement, subset, cardinality. Venn diagrams solve many counting problems, and the inclusion-exclusion principle generalises them: for two sets, the size of the union is the sum of the sizes minus the overlap.

Logic underlies proof. A statement and its contrapositive are equivalent; the converse and inverse are not. Proof by contradiction assumes the negation and derives a contradiction. Proof by induction establishes a base case and an inductive step, and is valid only over the natural numbers.

The commonest student error is proving an example rather than a general claim. Three cases support a conjecture but do not prove it; one counter-example refutes it.

Algebra, functions and equations

Quadratic equations are solved by factorisation, completing the square, or the formula. The discriminant determines the nature of the roots: positive for two real roots, zero for a repeated root, negative for complex roots. The sum and product of roots follow directly from the coefficients.

Functions map inputs to outputs; the domain is what is allowed, the range is what results. Composition and inverse functions require care about domains; an inverse exists only if the function is one-to-one.

Simultaneous equations can be solved by substitution, elimination or matrices. Word problems are solved by naming the unknowns, writing the equations from the conditions, checking the solution against the original wording - and rejecting solutions that do not fit the physical context.

> **Check against the question**
> A solution that satisfies the algebra but contradicts the physical situation - a negative length, a fractional number of students - is a signal that you set the equation up wrongly.

Sequences, series and counting

Arithmetic progressions add a constant difference; geometric progressions multiply by a constant ratio. The nth term and sum formulas follow from these definitions and are worth deriving once so they are not misremembered.

Permutations count arrangements where order matters; combinations count selections where it does not. The binomial theorem expands (a + b)^n using binomial coefficients, and Pascal's triangle generates them.

These tools are the foundation of probability and statistics, which is why they are taught early. A probability is the count of favourable outcomes over the count of equally likely outcomes, and the counting skills determine whether you can find either number.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. A set counting problem

In a class of 40, 25 study Mathematics, 20 study Physics, and 10 study both. How many study neither?

1. Let M and P be the sets; write the given values.
2. Apply inclusion-exclusion: |M union P| = 25 + 20 - 10 = 35.
3. Students studying at least one subject: 35.
4. Students studying neither: 40 - 35 = 5.
5. Check with a Venn diagram: only M = 15, only P = 10, both = 10, total 35.

> **What to carry forward**
> Draw the diagram after the formula. The two should agree; if they do not, you have misread the question.

2. Solving a quadratic with a condition

Find k if x squared + kx + 9 = 0 has equal roots.

1. Equal roots means the discriminant is zero.
2. Write the discriminant: b squared - 4ac = k squared - 36.
3. Set to zero: k squared = 36.
4. Solve: k = 6 or k = -6.
5. Check both by substitution - both give a perfect square.
6. State both answers; rejecting one without reason is an error.

> **What to carry forward**
> Translate the condition into an equation about the coefficients, then solve it.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Prove by induction that the sum of the first n natural numbers is n(n+1)/2.

15 marks

1. State the proposition for all n greater than or equal to 1.
2. Base case: check n = 1 on both sides.
3. Assume true for n = k.
4. Show it follows for n = k + 1 by adding the next term and simplifying.
5. Conclude by the principle of induction.

> **Where marks are lost**
> Omitting the base case or failing to state the induction hypothesis clearly.

Q2. Distinguish permutations from combinations with one worked example each.

10 marks

1. Define both in terms of whether order matters.
2. Give formulas and notation.
3. Work one arrangement and one selection example with the same numbers.
4. Explain why the combination result is smaller.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Set. A collection of distinct objects considered as a whole.

Union. The set of elements in either of two sets.

Intersection. The set of elements in both of two sets.

Inclusion-exclusion. A counting principle adding sizes and subtracting overlaps.

Contrapositive. The equivalent form of a conditional obtained by negating and reversing.

Discriminant. b squared minus 4ac, determining the nature of quadratic roots.

Domain. The set of inputs for which a function is defined.

Range. The set of outputs a function actually produces.

Arithmetic progression. A sequence with a constant difference between terms.

Geometric progression. A sequence with a constant ratio between terms.

Permutation. An arrangement in which order matters.

Combination. A selection in which order does not matter.

Your four-week plan

Week 1 - Sets

Counting.

- Solve five Venn diagram problems.
- Apply inclusion-exclusion to three cases.
- Practise set notation.

Week 2 - Algebra

Equations.

- Solve ten quadratics by three methods.
- Use the discriminant to classify roots.
- Set up three word problems.

Week 3 - Functions

Mapping.

- Find domains and ranges.
- Compose and invert functions.
- Solve simultaneous equations by two methods.

Week 4 - Sequences

Patterns and counting.

- Derive the nth term and sum for both progressions.
- Solve three permutation problems.
- Prove one statement by induction.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any elementary mathematics text covering sets, algebra and sequences.
- A problem-solving workbook with worked solutions.
- A short introduction to proof techniques for first-year students.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Solve x^2 - 5x + 6 = 0 and state the sum and product of the roots.

Answer: Factorising gives (x - 2)(x - 3) = 0, so x = 2 or 3. Sum of roots is 5 and product is 6.

2. Find the 10th term of the AP 3, 7, 11, ...

Answer: a = 3, d = 4, so the 10th term is 3 + 9(4) = 39.

3. Find the sum to infinity of 8, 4, 2, ...

Answer: a = 8, r = 0.5, so sum = 8/(1 - 0.5) = 16.

4. In a triangle, b = 7, c = 9 and angle A = 60 degrees. Find a.

Answer: a^2 = 49 + 81 - 2(7)(9)(0.5) = 130 - 63 = 67, so a is about 8.19.

5. Simplify 1/(sqrt(3) - 1) by rationalising.

Answer: Multiply numerator and denominator by sqrt(3) + 1 to get (sqrt(3) + 1)/2.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write out the factorisation identities and solve ten expressions using them.
- [ ] Derive the quadratic formula by completing the square.
- [ ] Draw the unit circle and mark the signs of sine, cosine and tangent in each quadrant.
- [ ] Solve five triangle problems using the sine and cosine rules.
- [ ] Work through three AP and three GP problems including sum to infinity.
- [ ] Do a timed 30-question arithmetic-and-algebra drill.
- [ ] Review every wrong answer and name the specific rule you misapplied.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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